

Does conditioning leave a structural memory?

Large-area structural evidence for the dislocation picture of conditioning

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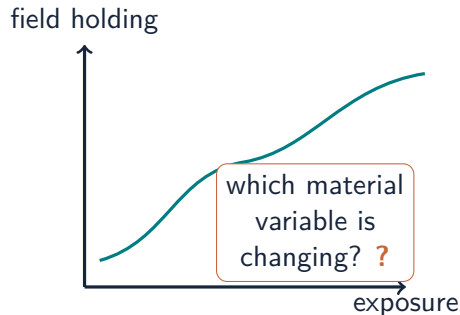
I. Popov (HUJI Nano) · V. M. Bjelland, W. Wuensch, L. Millar, S. Calatroni (CERN)



Conditioning works. What does it change?



- ▶ Conditioning is how every high-gradient device reaches its usable field.
- ▶ Breakdown rate has an extreme field dependence, $\sim E^{30}$.
- ▶ It tracks the number of **high-field pulses**, not the number of breakdowns.
- ▶ The field itself drives conditioning, not the arcs.



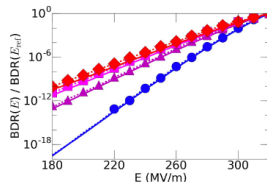
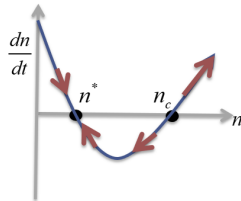
Something in the metal stores the improvement. We do not know what.

Source: Wuensch et al., Rep. Prog. Phys. (2026); Degiovanni et al., PRAB (2016).

Our model: conditioning as collective dislocation dynamics



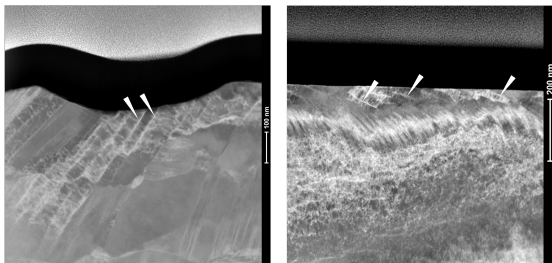
- ▶ Field holding rises FCC→BCC→HCP — the order of barriers to dislocation motion.
- ▶ The **MDDF** model: breakdown nucleates when the mobile-dislocation population, driven by Maxwell stress, runs away.
- ▶ One model gets the $\sim E^{30}$ rate, its temperature dependence, and the dark-current spikes.



Central prediction: conditioning *evolves the near-surface dislocation structure*.

Source: Engelberg et al., PRL 120, 124801 (2018); PRAB 22, 083501 (2019); PRAB 23, 123501 (2020).

STEM already saw it, at single spots



Cross-sectional STEM of conditioned hard (as-machined)

Cu: dislocation-denuded zone in the top ~ 200 nm.

- ▶ First **direct** structural evidence of field-driven near-surface plasticity.
- ▶ Seen at both 300 K and 30 K.
- ▶ Fits the clue that soft Cu conditions far more slowly than hard Cu: the starting population matters.
- ▶ FIB preparation plants its own dislocations; trust only large-scale structure.
- ▶ One cross-section cannot speak for the whole electrode.

Confirmed at one spot. The large-area comparison was still missing.

Source: Jacewicz et al., J. Appl. Phys. 137, 193302 (2025) — Uppsala/FREIA; Popov, HUJI. Korsbäck et al., PRAB 23, 033102 (2020). Shown at MeVArc 2025.

A tiny stress, repeated a billion times

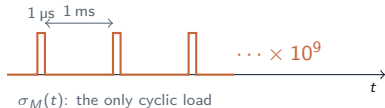


Maxwell stress

$$\sigma_M = \frac{\epsilon_0 E^2}{2}$$

at 80 MV/m: **0.028 MPa**

$\sim 10^3 \times$ below the yield of annealed Cu



- ▶ Fatigue at 10^8 – 10^{10} cycles rearranges dislocations irreversibly, well below yield.
- ▶ Here: $\sim 10^9$ conditioning pulses.
- ▶ Pulsed DC means **no RF pulsed heating** and no bulk currents. Maxwell stress is the only cyclic load.
- ▶ MDDF is **threshold-free**: no minimum stress amplitude.

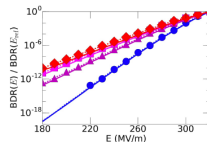
The model predicts a large-area signature. Last year we asked for it; this year we looked.

Source: Mughrabi, MMTB 40, 431 (2009); Stanzl-Tschegg et al., Int. J. Fatigue 29, 2050 (2007).

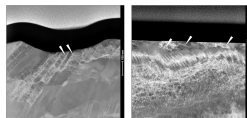
Last year we asked for more forensics. Here it is.



MeVArc 2025



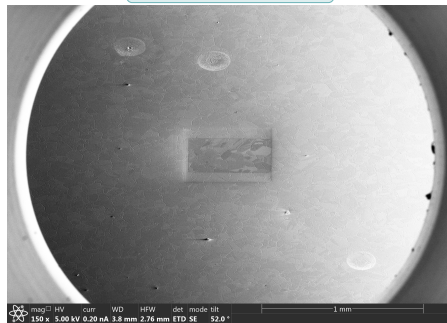
MDDF model



local STEM denuded
zone

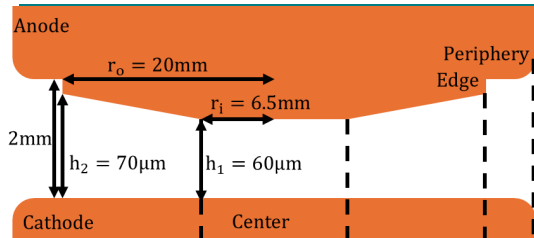
Closed with: other materials? more samples?

MeVArc 2026

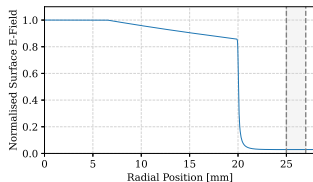


Large-area EBSD across one cathode

One cathode turns field exposure into a controlled coordinate



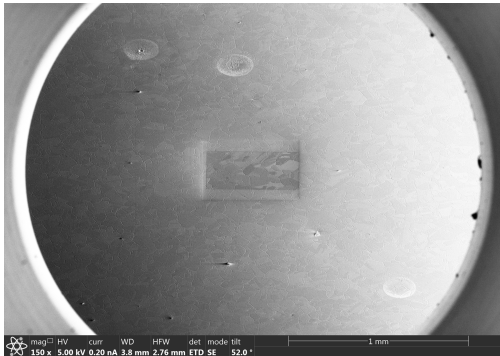
- ▶ Heat-treated OFE Cu, pulsed DC
- ▶ 1 μ s pulses at 1 kHz
- ▶ Peak field ~ 80 MV/m at center
- ▶ Nine EBSD regions: center, edge, periphery, and external reference



The field gradient gives us internal controls on a single specimen.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192; geometry from Bjelland et al.

EBSD maps plastic activity far from breakdown craters



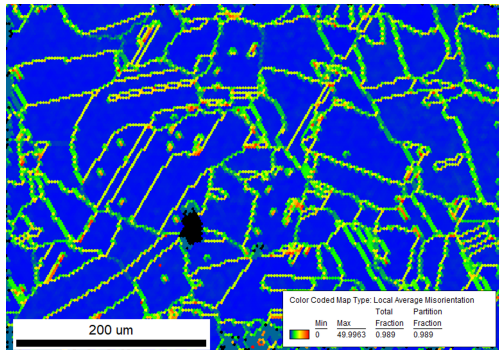
Cleaned ROI is hundreds of microns from visible craters.

- ▶ Large area where TEM could only sample a point
- ▶ 500 μm -wide maps; 3 μm step (lateral only); information depth a few tens of nm
- ▶ $\sim 24,000$ indexed points per ROI
- ▶ LAM, KAM, and LOS probe intragrain orientation gradients
- ▶ Misorientation is a proxy for GND density

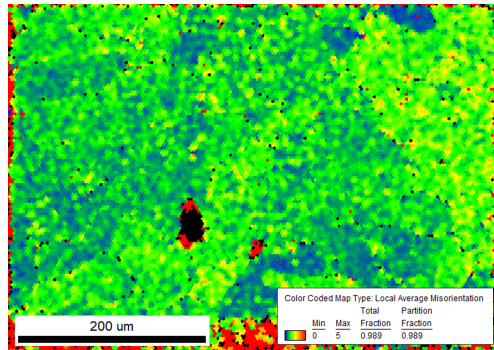
We measure the cumulative response to the field, not crater damage.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192; information depth after Chen et al., Ultramicroscopy 111, 1488 (2011).

How to read a LAM map: wide-angle vs low-angle



0–50°: grain boundaries dominate

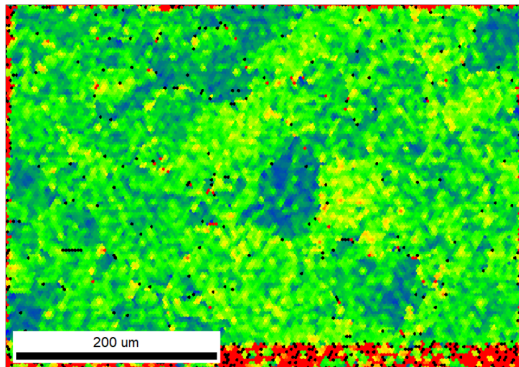


0–5° scale: intragrain curvature

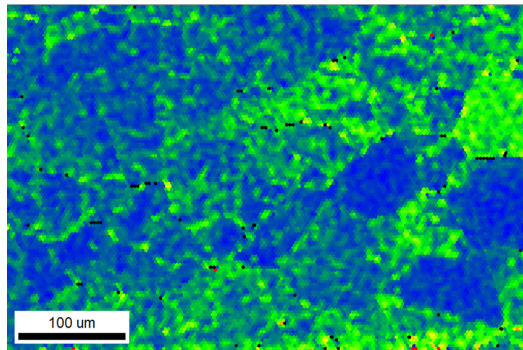
Rescaling to 0–5° exposes the intragrain misorientation we care about.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192. Same area, two color scales.

High-field copper contains more intragrain curvature



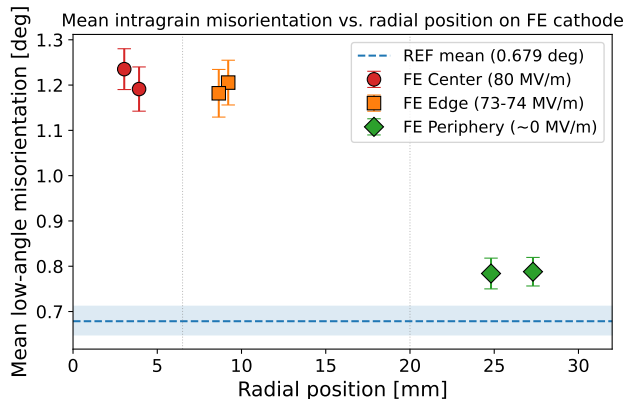
Field-exposed center



Unexposed reference

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192. Same LAM color scale.

The signal separates into three exposure tiers

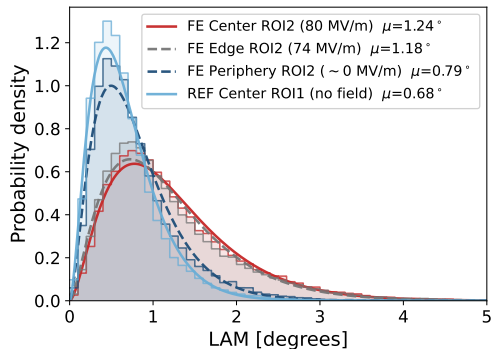


- ▶ Center and edge: $\sim 1.2^\circ$
- ▶ External reference: $\sim 0.68^\circ$
- ▶ High field is $\sim 75\%$ higher
- ▶ Grain size does not follow this hierarchy

The ordering follows field history, not sample position or grain size.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192.

Field exposure builds a high-misorientation tail



LAM tail above 2°

~ 0.14 field-exposed

~ 0.016 reference

$\sim 8\times$ **increase**

Gamma shape $k \approx 2.7$ fixed; the
scale roughly doubles.

The whole distribution stretches; a uniform offset cannot do that.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192.

The large-area test confirms the model's prediction



- ▶ First **large-area** confirmation that conditioning evolves the near-surface dislocation structure, as MDDF predicted.
- ▶ No conflict with the 2025 denuded zone: one mechanism, two starting populations (hard Cu depletes; heat-treated Cu builds curvature).
- ▶ The three-tier ordering also tracks the conditioning variable E_S : a *candidate* structural basis for it.

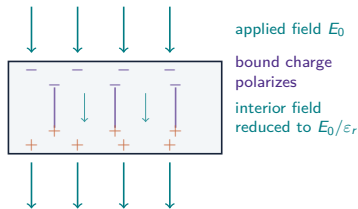
Before, dislocations were plausible. Now, a dislocation-sensitive state maps the conditioning history.

Source: Ashkenazy et al., PRAB, under revision; arXiv:2606.19192.

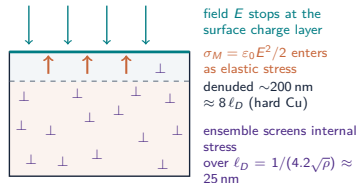
Elastic screening: dislocations respond like bound charge



Dielectric in an applied field



Copper under the Maxwell traction



applied field $\leftrightarrow$ Maxwell traction · bound charge $\leftrightarrow$ dislocations (dipole regime) · Debye length $\leftrightarrow \ell_D$ · screened interior field $\leftrightarrow$ screened internal stress of the depleted layer

2D single-slip theory at $\rho \sim 10^{14} \text{ m}^{-2}$; no theory yet treats cyclic loading with a free surface, so the $\sim 200 \text{ nm}$ thickness is observed, not predicted.

Charge screens the field at the surface. Dislocations screen the internal stress of a depleted layer, over $\sim 25 \text{ nm}$.

Source: Groma et al., PRL 96, 165503 (2006); Lemaître et al., PRE 104, 024904 (2021); Livne et al., PRE 107, 055004 (2023); PRAB, under revision; arXiv:2606.19192.

Copper establishes the effect, not universality



What is established

- ▶ Large-area structural difference
- ▶ Three independent metrics
- ▶ Internal and external controls
- ▶ Confirms the MDDF prediction

What remains open

- ▶ One conditioned cathode, one experiment
- ▶ Full depth profile
- ▶ Grain-orientation dependence
- ▶ More conditioned cathodes
- ▶ Cross-material generality

Call for cooperation: more cathodes, other materials, other facilities — let's compare notes.

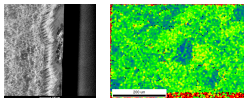
A stronger test changes both the material and the loading history.

Next: go deeper, and change the material



Depth

- ▶ Matched high-field and reference FIB lamellae; TEM/STEM + EBSD through the top $\sim 1\ \mu\text{m}$
- ▶ Artifact controls from the 2025 “red herrings”
- ▶ One depth profile from STEM depletion to EBSD curvature?



Material & loading

- ▶ Stainless steel: lower stacking-fault energy, different dislocation mobility and slip
- ▶ Continuous DC instead of pulsed loading
- ▶ Possible metastable phase transformation



Depth continues the Uppsala STEM line; material tests the mechanism itself.

Source: Interim work plan (2026); RFX project plan.

Stainless steel: a test chosen to be hard on the model



Copper

FCC, higher SFE
pulsed DC, $\sim 10^9$ cycles

AISI 304L SS

metastable austenite, lower SFE
continuous DC, possible $\gamma \rightarrow \alpha'$

Design: apex-to-periphery field gradient; two exposed + two matched references.

Any of the three outcomes constrains the mechanism; the test is not pass-or-fail.

SS tracks field

cross-material structural state

SS differs

material-specific carriers

no resolved change

bounds dose and sensitivity

Status: 90 h and 73 h DC, peak 59–61 MV/m, no full breakdowns; EBSD indexes three of four samples, one field-exposed sample still does not.



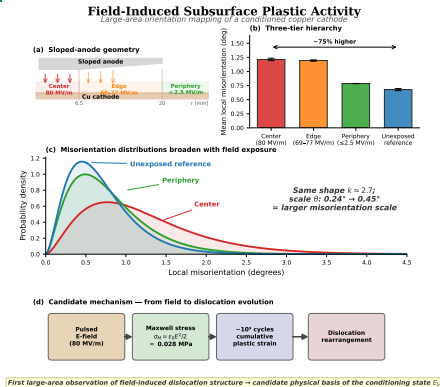
Consorzio RFX — De Lorenzi, Pilan, Spada

Source: RFX electrode geometry and plan; project record through 15 Sep 2026; first-look data in backup.

Conditioning appears to leave a subsurface memory



1. Our model predicts conditioning evolves the near-surface dislocation structure.
2. 2025 STEM saw it locally; **large-area EBSD now confirms it** ($\sim 75\%$ higher misorientation).
3. Depth and stainless steel test whether the memory survives a change of scale, alloy, and loading.



Last year: “other materials? more samples?” — here is the large-area answer.

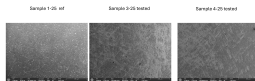
Copper result: PRAB, under revision; arXiv:2606.19192.

Backup: RFX first look (preliminary)

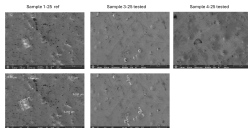


Preliminary — no field effect claimed

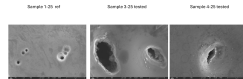
- ▶ AISI 304L at RFX HVPTF; continuous DC: 90 h (#3-25), 73 h (#4-25); peak 59–61 MV/m; no full breakdowns
- ▶ SEM: “exploded” near-surface inclusions (3–7 μm) on field-exposed samples; reference cavities round and clean
- ▶ EBSD: #1-25, #2-25, #3-25 index (97–99.5%); #4-25 patterns poor, not repolished
- ▶ Apex-vs-side contrast reproduces on never-installed #2-25: forming geography



Ref vs field-exposed overview



Sectioned electrode



Source: RFX project record through 15 Sep 2026; SEM report, Inna Popov (20 May 2026).

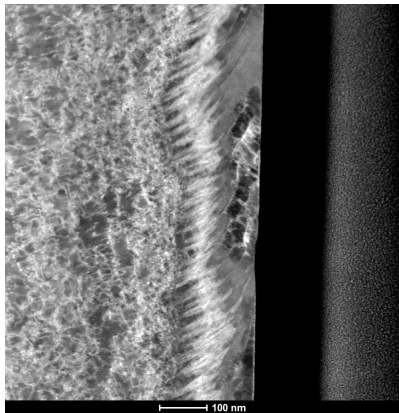
Backup: RFX sample and data status (15 Sep 2026)



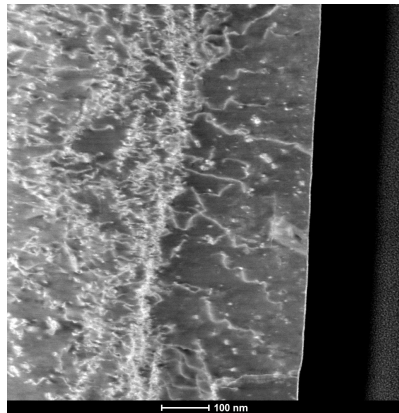
Sample	Role	Exposure	Status
#1-25	reference	none	SEM; EBSD apex + side, 97% indexed
#2-25	reference	none	EBSD apex (18 Aug, 3 maps) + side; LAM done
#3-25	field-exposed	90 h DC, peak 61 MV/m	SEM; EBSD apex + side, 99.5% indexed; LAM done
#4-25	field-exposed	73 h DC, peak 59 MV/m	SEM/EDS; EBSD patterns poor, not repolished

- ▶ Received since July: conditioning report and $E(\theta)$; alloy confirmed as 304L, mill certificate unrecoverable; electropolishing partly documented; five per-pixel maps.
- ▶ Still open: raw patterns unavailable; per-pixel data for the wide-grain class; matched EDS overlay checks.

Backup: field-exposed vs reference STEM (2025)



Field-exposed: denuded near surface



Reference: dislocation-rich to surface

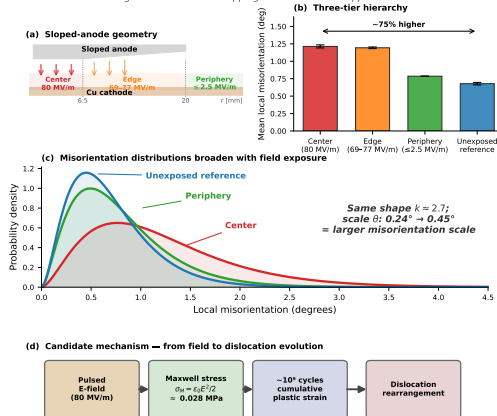
Source: Jacewicz et al., J. Appl. Phys. 137, 193302 (2025).

Backup: copper result in one figure



Field-Induced Subsurface Plastic Activity

Large-area orientation mapping of a conditioned copper cathode



First large-area observation of field-induced dislocation structure → candidate physical basis of the conditioning state E_s